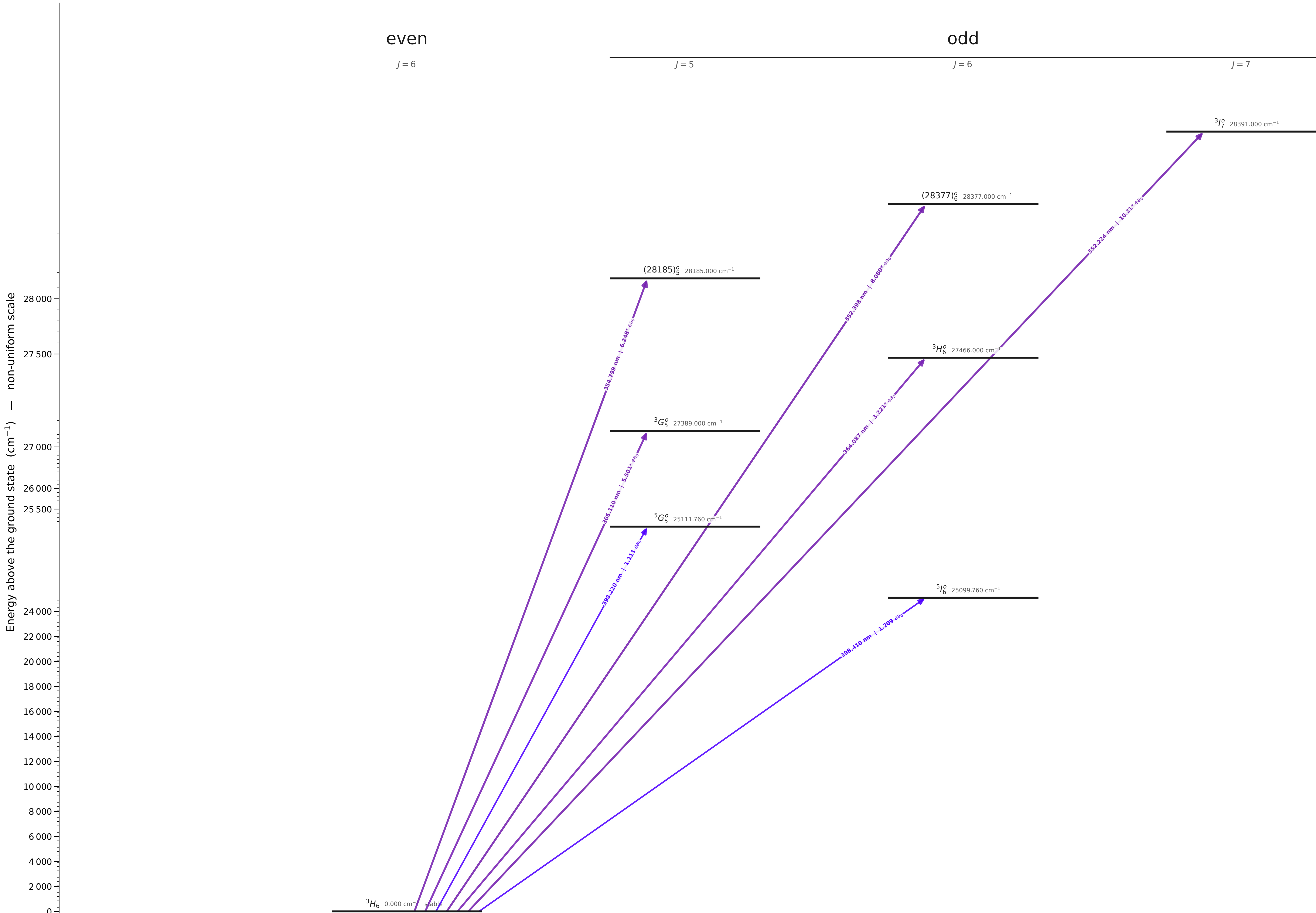


Fermium

Fm (**Fm I**). Classified transitions below 2 μm (levels and lines from the literature): 8 levels, 7 lines.

Fm⁺ ionisation limit 52400.00 cm⁻¹ = 6.49677 eV (190.840 nm)



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	⟨J er J'⟩ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
³ H ₆ - ³ I ₇	352.2243	352.1236	851.1408	10.21*	3.223e+08	-	-	ground-state resonance found in the 2005 level search
³ H ₆ - (28377) ₆	352.3981	352.2973	850.7211	8.080*	2.325e+08	-	-	ground-state resonance found in the 2005 level search
³ H ₆ - (28185) ₅	354.7987	354.6973	844.9650	6.248*	1.610e+08	-	-	ground-state resonance found in the 2005 level search
³ H ₆ - ³ H ₆	364.0865	363.9828	823.4100	3.221*	3.351e+07	-	-	ground-state resonance found in the 2005 level search
³ H ₆ - ³ G ₅	365.1101	365.0061	821.1016	5.501*	1.145e+08	-	-	ground-state resonance found in the 2005 level search
³ H ₆ - ⁵ G ₅	398.2198	398.1072	752.8316	1.111	3.600e+06	-	-	resonance R2: first step of two-step resonance ionization...
³ H ₆ - ⁵ I ₆	398.4102	398.2975	752.4719	1.209	3.600e+06	-	-	resonance R1: first step of two-step resonance ionization...

λ, ν from the level energies of the cited literature (NIST lists no excited levels) unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

²⁵⁵Fm hyperfine structure (*I* = 7/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)									
³ H ₆	-149.5(11)	-10460(40)	F=5/2: -497.48	F=7/2: +762.23	F=9/2: +1974.32	F=11/2: +2833.14	F=13/2: +2965.12	F=15/2: +1928.77	F=17/2: -785.32	F=19/2: -5754.50		
⁵ I ₆	-309.2(12)	-12790(40)	F=5/2: +2488.47	F=7/2: +3586.38	F=9/2: +4499.67	F=11/2: +4854.61	F=13/2: +4194.40	F=15/2: +1979.22	F=17/2: -2413.84	F=19/2: -9690.70		
⁵ G ₅	-9.7(13)	-13570(50)	F=3/2: -5676.63	F=5/2: -3600.76	F=7/2: -1146.88	F=9/2: +1232.68	F=11/2: +2956.36	F=13/2: +3313.33	F=15/2: +1463.56	F=17/2: -3562.25		

Measured A, B constants; sources in the data file.

Arrow label: vacuum wavelength | reduced dipole matrix element |⟨J||er||J'⟩| in ea₀

Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model or semi-empirical calculation.

Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, *n* = 70.

Level caption: energy (from the cited literature; NIST lists no excited levels) and measured lifetime (where one exists).

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
³ H ₆ - ⁵ G ₅	398.220	245-250	30400(2200)
³ H ₆ - ⁵ G ₅	398.220	246-250	21000(1700)
³ H ₆ - ⁵ G ₅	398.220	248-250	9500(1000)
³ H ₆ - ⁵ G ₅	398.220	249-250	5800(3400)
³ H ₆ - ⁵ G ₅	398.220	254-250	-27000(1400)
³ H ₆ - ⁵ G ₅	398.220	255-250	-31500(900)
³ H ₆ - ⁵ G ₅	398.220	257-250	-39500(2400)

Shift = ν(first isotope) - ν(second isotope).